

SI 140A-01 Probability & Statistics for EECS, Fall 2026

Homework 2

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Read all the instructions below carefully before you start working on the assignment, and before you make a submission.

- You are required to write down all the major steps towards making your conclusions; otherwise you may obtain limited points of the problem.
- Write your homework in English; otherwise you will get no points of this homework.
- Any form of plagiarism will lead to 0 point of this homework.

Problem 1

Show that for any events A and B ,

$$P(A) + P(B) - 1 \leq P(A \cap B) \leq P(A \cup B) \leq P(A) + P(B).$$

For each of these three inequalities, give a simple criterion for when the inequality is actually an equality (e.g., give a simple condition such that $P(A \cap B) = P(A \cup B)$ if and only if the condition holds).

Solution:

First:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

So:

$$P(A \cap B) = P(A) + P(B) - P(A \cup B)$$

In addition:

$$P(A \cup B) \leq 1$$

Therefore:

$$P(A) + P(B) - 1 \leq P(A \cap B)$$

The left inequality is proven, and the two parts are equal if and only if $P(A \cup B) = 1$.

Now consider the middle part:

$$A \cap B \subseteq A \cup B$$

So:

$$P(A \cap B) \leq P(A \cup B)$$

The two sides are equal if and only if $P(A \cap B) = P(A) = P(B)$.

Consider the right part:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

So:

$$P(A) + P(B) = P(A \cup B) + P(A \cap B)$$

Therefore:

$$P(A \cup B) \leq P(A) + P(B)$$

The two parts are equal if and only if $P(A \cap B) = 0$.

Problem 2

Let A and B be events. The *symmetric difference* $A \triangle B$ is defined to be the set of all elements that are in A or B but not both. In logic and engineering, this event is also called the XOR (*exclusive or*) of A and B . Show that

$$P(A \triangle B) = P(A) + P(B) - 2P(A \cap B),$$

directly using the axioms of probability.

Solution:

According to the definition, $A \triangle B$ represents the set that includes the elements in A or in B but not in $A \cap B$. So:

$$A \triangle B = (A \setminus B) \cup (B \setminus A)$$

$A \setminus B$ and $B \setminus A$ are mutually exclusive:

$$(A \setminus B) \cap (B \setminus A) = \emptyset$$

According to the third axiom:

$$P[(A \setminus B) \cup (B \setminus A)] = P(A \triangle B) = P(A \setminus B) + P(B \setminus A)$$

Since $A = (A \cap B) \cup (A \setminus B)$ with the two parts disjoint, the third axiom gives $P(A) = P(A \cap B) + P(A \setminus B)$, hence $P(A \setminus B) = P(A) - P(A \cap B)$. Similarly $P(B \setminus A) = P(B) - P(A \cap B)$.

So:

$$P(A \triangle B) = P(A) + P(B) - 2P(A \cap B)$$

The equation is proven.

Problem 3

Events A and B are *independent* if $P(A \cap B) = P(A)P(B)$.

1. Give an example of independent events A and B in a finite sample space S (with neither equal to $\emptyset$ or S).
2. Consider the experiment of picking a random point in the rectangle

$$R = \{(x, y) : 0 < x < 1, 0 < y < 1\},$$

where the probability of the point being in any particular region contained within R is the area of that region. Let A_1 and B_1 be rectangles contained within R , with areas not equal to 0 or 1. Let A be the event that the random point is in A_1 , and B be the event that the random point is in B_1 . Give a geometric description of when it is true that A and B are independent. Also, give an example where they are independent and another example where they are not independent.

3. Show that if A and B are independent, then

$$P(A \cup B) = P(A) + P(B) - P(A)P(B) = 1 - P(A^c)P(B^c).$$

Solution:

1. Consider tossing a fair coin twice. It's obvious that S is:

$$S = \{HH, HT, TH, TT\}$$

Let event A represents the situation that the first time's result is head ($A = \{HH, HT\}$). And let event B represent the situation that the second time's result is head ($B = \{TH, HH\}$):

$$P(A \cap B) = P(\{HH\}) = \frac{1}{4}$$

$$P(A)P(B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

So:

$$P(A \cap B) = P(A)P(B)$$

Events A and B are independent.

2. Let $Area(X)$ represents the area of x . It's obvious that $Area(R) = 1$. According to the question, $Area(X) = P(X)$. A and B are independent if and only if:

$$Area(A_1 \cap B_1) = Area(A_1)Area(B_1)$$

Geometrically, A and B are independent exactly when B_1 occupies the same proportion of A_1 as it does of R .

And here are the examples for independent and non-independent:

- (1). Let $A_1 = \{(x, y) : 0 < x < \frac{1}{2}, 0 < y < 1\}$ and $B_1 = \{(x, y) : 0 < x < 1, 0 < y < \frac{1}{2}\}$.

$$Area(A_1) = \frac{1}{2}$$

$$Area(B_1) = \frac{1}{2}$$

$$Area(A_1 \cap B_1) = \frac{1}{4}$$

$$Area(A_1)Area(B_1) = Area(A_1 \cap B_1)$$

So A and B are independent.

- (2). Let:

$$A_1 = \{(x, y) : 0 < x < \frac{1}{2}, 0 < y < 1\}$$

$$B_1 = \{(x, y) : \frac{1}{2} < x < 1, 0 < y < 1\}$$

$Area(A_1 \cap B_1) = 0$ but $Area(A_1)Area(B_1) = \frac{1}{4} \neq 0$.

So A and B are not independent.

3. Because A and B are independent.

So:

$$P(A)P(B) = P(A \cap B)$$

Therefore:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = P(A) + P(B) - P(A)P(B)$$

In addition, A^C and B^C are independent, So:

$$P(A^C)P(B^C) = P(A^C \cap B^C)$$

According to De Morgan's law:

$$P(A^C \cap B^C) = P((A \cup B)^C) = P(A^C)P(B^C)$$

Therefore:

$$1 - P(A^C)P(B^C) = 1 - P((A \cup B)^C) = P(A \cup B)$$

So:

$$P(A \cup B) = P(A) + P(B) - P(A)P(B) = 1 - P(A^C)P(B^C).$$

The formula is proven.

Problem 4

Consider four nonstandard dice (the *Efron dice*), whose sides are labeled as follows (the 6 sides on each die are equally likely).

A: 4, 4, 4, 4, 0, 0

B: 3, 3, 3, 3, 3, 3

C: 6, 6, 2, 2, 2, 2

D: 5, 5, 5, 1, 1, 1

These four dice are each rolled once. Let A be the result for die A , B be the result for die B , etc.

1. Find $P(A > B)$, $P(B > C)$, $P(C > D)$, and $P(D > A)$.
2. Is the event $A > B$ independent of the event $B > C$? Is the event $B > C$ independent of the event $C > D$? Explain.

Solution:

1.

$$P(A > B) = \frac{2}{3}$$

$$P(B > C) = \frac{2}{3}$$

$$P(C > D) = \frac{1}{3} + \frac{2}{3} \cdot \frac{1}{2} = \frac{2}{3}$$

$$P(D > A) = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \frac{1}{3} = \frac{2}{3}$$

2.

$$P((A > B) \cap (B > C)) = \frac{2}{3} \cdot \frac{2}{3} = \frac{4}{9}$$

$$P((B > C) \cap (C > D)) = \frac{2}{3} \cdot \frac{1}{2} = \frac{1}{3}$$

$$P(A > B)P(B > C) = \frac{4}{9}$$

$$P(B > C)P(C > D) = \frac{4}{9} \neq \frac{1}{3}$$

Therefore:

The events $A > B$ and $B > C$ are independent, since $B \equiv 3$ implies they depend only on the independent dice A and C . The events $B > C$ and $C > D$ are not independent, since both depend on die C .